ABL PYTHON PROGRAMMING LABORATORY MANUAL

ACTIVITY-BASED LEARNING COURSE CODE & EXECUTION WORKBOOK

Course Title	Activity-Based Learning: Python Programming	**Document Code**	ABL-PY-2026-MANUAL
Target System	Python 3.x Interpreter / IDLE / Jupyter	**Standard Format**	Aim | Algorithm | Pseudo Code | Code | Output | Conclusion

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EXPERIMENT 01: Arithmetic Operations & Data Input **LAB SEC: ABL-01**

1. AIM

To write and execute a Python program that accepts two numeric inputs from the user and performs basic arithmetic operations (Addition, Subtraction, Multiplication, Division, and Modulus).

2. ALGORITHM

1. Start the program.
2. Prompt the user to enter two numbers, `num1` and `num2`.
3. Compute addition: `sum = num1 + num2`.
4. Compute subtraction: `diff = num1 - num2`.
5. Compute multiplication: `prod = num1 * num2`.
6. Compute division: `div = num1 / num2` (check for zero divisor).
7. Compute modulus: `mod = num1 % num2`.
8. Display the calculated results on screen.
9. Stop the program.

3. PSEUDO CODE

```
BEGIN
   READ num1, num2
   SET add_res = num1 + num2
   SET sub_res = num1 - num2
   SET mul_res = num1 * num2
   IF num2 != 0 THEN
      SET div_res = num1 / num2
      SET mod_res = num1 % num2
   ELSE
      PRINT "Division by zero not allowed"
   ENDIF
   PRINT add_res, sub_res, mul_res, div_res, mod_res
END
```

4. PYTHON PROGRAM

```python
# Program to perform basic arithmetic operations
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print(f"Addition: {num1} + {num2} = {num1 + num2}")
print(f"Subtraction: {num1} - {num2} = {num1 - num2}")
print(f"Multiplication: {num1} * {num2} = {num1 * num2}")
if num2 != 0:
    print(f"Division: {num1} / {num2} = {num1 / num2:.2f}")
    print(f"Modulus: {num1} % {num2} = {num1 % num2}")
else:
    print("Error: Division by zero is undefined.")
```

5. SAMPLE OUTPUT

```
Enter first number: 25
Enter second number: 4
Addition: 25.0 + 4.0 = 29.0
Subtraction: 25.0 - 4.0 = 21.0
Multiplication: 25.0 * 4.0 = 100.0
Division: 25.0 / 4.0 = 6.25
Modulus: 25.0 % 4.0 = 1.0
```

6. CONCLUSION

The program was successfully executed. The basic arithmetic operations and dynamic dynamic type conversion using `float()` were demonstrated and verified.

EXPERIMENT 02: Prime Number Determination — LAB SEC: ABL-02

1. AIM

To design a Python program to check whether a given integer is a prime number using conditional statements and loops.

2. ALGORITHM

1. Start the program.
2. Read an integer n from the user.
3. If `n <= 1`, set flag as `False` (not prime).
4. Iterate a loop variable `i` from 2 up to `sqrt(n)`.
5. If `n % i == 0`, set flag as `False` and break the loop.
6. If loop completes without finding a factor, set flag as `True`.
7. Display prime status based on the flag value.
8. Stop the program.

3. PSEUDO CODE

```
BEGIN
   READ n
   IF n <= 1 THEN
      is_prime = FALSE
   ELSE
      is_prime = TRUE
      FOR i = 2 TO sqrt(n) DO
         IF n MOD i == 0 THEN
            is_prime = FALSE
            BREAK
         ENDIF
      ENDFOR
   ENDIF
   IF is_prime == TRUE THEN
      PRINT n, "is a Prime Number"
   ELSE
      PRINT n, "is Not a Prime Number"
   ENDIF
END
```

4. PYTHON PROGRAM

```
import math

n = int(input("Enter an integer: "))
is_prime = True

if n <= 1:
    is_prime = False
else:
    for i in range(2, int(math.isqrt(n)) + 1):
        if n % i == 0:
            is_prime = False
            break

if is_prime:
    print(f"{n} is a Prime Number.")
else:
    print(f"{n} is NOT a Prime Number.")
```

5. SAMPLE OUTPUT

```
Enter an integer: 29
29 is a Prime Number.
```

6. CONCLUSION

The logic for determining prime numbers was implemented using an efficient O(√N) iteration strategy and successfully validated with sample inputs.

EXPERIMENT 03: Fibonacci Sequence Generation — LAB SEC: ABL-03

1. AIM

To develop a Python program that generates and prints the Fibonacci series up to n terms using loop structures.

2. ALGORITHM

1. Start the program.
2. Input the number of terms n.
3. Initialize first two terms a = 0 and b = 1.
4. If n <= 0, display error message.
5. If n == 1, print a.
6. For terms from 1 to n, print a, calculate next = a + b, and update a = b, b = next.
7. Stop the program.

3. PSEUDO CODE

```
BEGIN
   READ n
   SET a = 0, b = 1
   IF n <= 0 THEN
      PRINT "Enter a positive integer"
   ELSE
      FOR count = 1 TO n DO
         PRINT a
         SET temp = a + b
         SET a = b
         SET b = temp
      ENDFOR
   ENDIF
END
```

4. PYTHON PROGRAM

```
n = int(input("Enter number of terms: "))
a, b = 0, 1

if n <= 0:
    print("Please enter a positive integer.")
elif n == 1:
    print("Fibonacci sequence:", a)
else:
    print("Fibonacci sequence:", end=" ")
    for _ in range(n):
        print(a, end=" ")
        a, b = b, a + b
    print()
```

5. SAMPLE OUTPUT

```
Enter number of terms: 8
Fibonacci sequence: 0 1 1 2 3 5 8 13
```

6. CONCLUSION

The Fibonacci sequence generator was created using dynamic variable updates in Python, illustrating iterative sequence construction.

EXPERIMENT 04: List Operations & Sorting — LAB SEC: ABL-04

1. AIM

To implement list manipulations in Python including insertion, element search, calculation of sum/average, and sorting.

2. ALGORITHM

1. Start the program.
2. Initialize an empty list or take space-separated inputs into a list.
3. Calculate total sum using `sum()` and length using `len()` to compute average.
4. Sort the list in ascending order using the `sort()` method.
5. Prompt for a search target element and apply linear search or index lookups.
6. Display the computed statistics, sorted array, and search results.
7. Stop the program.

3. PSEUDO CODE

```
BEGIN
   READ numbers_list
   SET total_sum = SUM(numbers_list)
   SET avg = total_sum / LENGTH(numbers_list)
   SORT numbers_list ASCENDING
   READ target
   IF target IN numbers_list THEN
      PRINT "Element found at index", INDEXOF(target)
   ELSE
      PRINT "Element not found"
   ENDIF
   PRINT total_sum, avg, numbers_list
END
```

4. PYTHON PROGRAM

```
# Program for list processing and search operations
data = [int(x) for x in input("Enter integers separated by space: ").split()]

total_sum = sum(data)
avg = total_sum / len(data) if data else 0

print(f"List: {data}")
print(f"Sum: {total_sum} | Average: {avg:.2f}")

data.sort()
print(f"Sorted List: {data}")

target = int(input("Enter element to search: "))
if target in data:
    print(f"Element {target} found at sorted index {data.index(target)}.")
else:
    print(f"Element {target} not found in list.")
```

5. SAMPLE OUTPUT

```
Enter integers separated by space: 42 12 89 23 7
List: [42, 12, 89, 23, 7]
Sum: 173 | Average: 34.60
Sorted List: [7, 12, 23, 42, 89]
Enter element to search: 23
Element 23 found at sorted index 2.
```

6. CONCLUSION

Python's built-in list methods (`sort`, `index`, list comprehensions) were leveraged to perform collection manipulations efficiently.

EXPERIMENT 05: Character Frequency Count using Dictionaries — LAB SEC: ABL-05

1. AIM

To construct a Python program that accepts a string input and computes the occurrence frequency of each unique character using dictionary data structures.

2. ALGORITHM

1. Start the program.
2. Input a string `text` from the user.
3. Initialize an empty dictionary `freq_map`.
4. Iterate through each character `ch` in `text`.
5. If `ch` exists in `freq_map`, increment its count by 1; else set count to 1.
6. Display the key-value pairs representing character counts.
7. Stop the program.

3. PSEUDO CODE

```
BEGIN
   READ text
   SET freq_dict = EMPTY_DICTIONARY
   FOR EACH char IN text DO
      IF char IN freq_dict THEN
         freq_dict[char] = freq_dict[char] + 1
      ELSE
         freq_dict[char] = 1
      ENDIF
   ENDFOR
   PRINT freq_dict
END
```

4. PYTHON PROGRAM

```
text = input("Enter a string: ")
freq_map = {}

for char in text:
    freq_map[char] = freq_map.get(char, 0) + 1

print("\nCharacter Frequencies:")
for char, count in freq_map.items():
    print(f"  '{char}': {count}")
```

5. SAMPLE OUTPUT

```
Enter a string: abl python
Character Frequencies:
  'a': 1
  'b': 1
  'l': 1
  ' ': 1
  'p': 1
  'y': 1
  't': 1
  'h': 1
  'o': 1
  'n': 1
```

6. CONCLUSION

Dictionary key-value mapping was utilized to achieve an O(N) frequency calculation for arbitrary input strings.

EXPERIMENT 06: File I/O & Word Count Analysis — LAB SEC: ABL-06

1. AIM

To write a Python program that creates a text file, writes content into it, reads the file back, and counts the total number of lines, words, and characters.

2. ALGORITHM

1. Start the program.
2. Open a target file in write mode (`'w'`) and write sample multi-line text.
3. Close the file writer.
4. Open the file in read mode (`'r'`).
5. Initialize counters: `line_cnt = 0`, `word_cnt = 0`, `char_cnt = 0`.
6. Iterate line by line: increment line count, add line length to char count, split line into words, and add word count.
7. Display summary analysis metrics.
8. Stop the program.

3. PSEUDO CODE

```
BEGIN
   OPEN "sample.txt" FOR WRITING
   WRITE lines of text
   CLOSE file

   OPEN "sample.txt" FOR READING
   SET line_count = 0, word_count = 0, char_count = 0
   FOR EACH line IN file DO
      line_count = line_count + 1
      char_count = char_count + LENGTH(line)
      words = SPLIT(line)
      word_count = word_count + LENGTH(words)
   ENDFOR
   PRINT line_count, word_count, char_count
END
```

4. PYTHON PROGRAM

```
filename = "abl_sample.txt"

# Write sample content
with open(filename, "w") as f:
    f.write("Welcome to ABL Python Lab.\nPython is concise and readable.\nHappy
Coding!\n")

# Read and analyze content
line_cnt = word_cnt = char_cnt = 0

with open(filename, "r") as f:
    for line in f:
        line_cnt += 1
        char_cnt += len(line)
        word_cnt += len(line.split())

print(f"File '{filename}' Metrics:")
print(f"  Total Lines      : {line_cnt}")
print(f"  Total Words      : {word_cnt}")
print(f"  Total Characters : {char_cnt}")
```

5. SAMPLE OUTPUT

```
File 'abl_sample.txt' Metrics:
  Total Lines      : 3
  Total Words      : 11
  Total Characters : 71
```

6. CONCLUSION

File handling mechanics (context managers, safe I/O operations) and string parsing techniques were verified successfully.